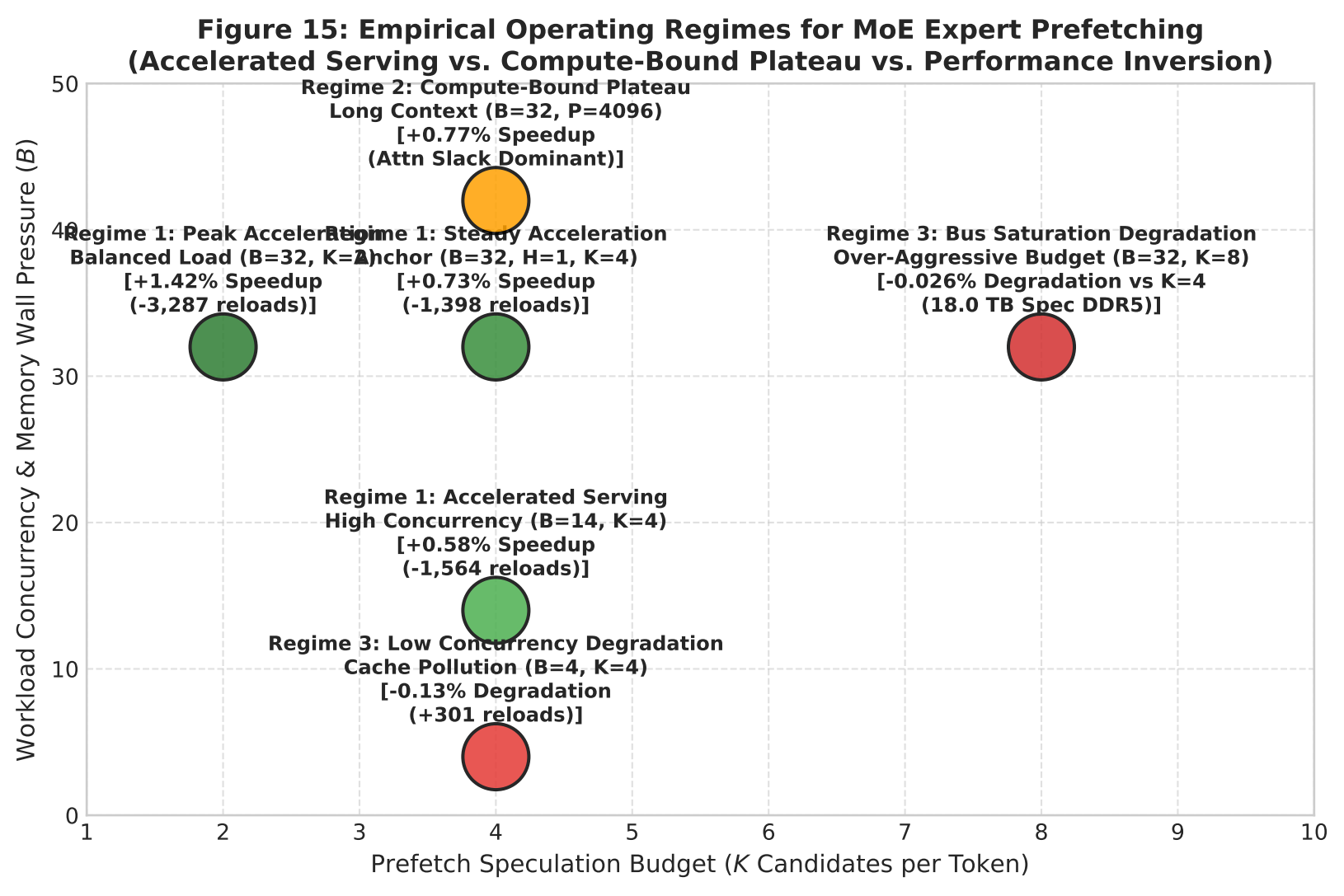
Figure 15: Empirical Operating Regimes for MoE Expert Prefetching
(Accelerated Serving vs. Compute-Bound Plateau vs. Performance Inversion)
Regime 2: Compute-Bound Plateau
Long Context (B=32, P=4096)
[+0.77% Speedup
(Attn Slack Dominant)]
Regime 1: Peak Acceleration
Balanced Load (B=32, K=2)
[+1.42% Speedup
(-3,287 reloads)]
Regime 1: Steady Acceleration
Anchor (B=32, H=1, K=4)
[+0.73% Speedup
(-1,398 reloads)]
Regime 3: Bus Saturation Degradation
Over-Aggressive Budget (B=32, K=8)
[-0.026% Degradation vs K=4
(18.0 TB Spec DDR5)]
Regime 1: Accelerated Serving
High Concurrency (B=14, K=4)
[+0.58% Speedup
(-1,564 reloads)]
Regime 3: Low Concurrency Degradation
Cache Pollution (B=4, K=4)
[-0.13% Degradation
(+301 reloads)]
Workload Concurrency & Memory Wall Pressure (B)
Prefetch Speculation Budget (K Candidates per Token)
0
10
20
30
40
50
1
2
3
4
5
6
7
8
9
10